Why Fewest-Moves Methods Work: A Group-Theoretic Account of Reduction and Its Effectiveness

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Abstract

Competitive Fewest-Moves (FMC) solvers do not search the 4.3×10^{19}-state cube group directly; they *reduce*, walking the cube down a chain of nested subgroups

$$G = G_0 \supset G_1 \supset \cdots \supset G_k = \{e\},$$

each stage "killing" one feature (edge orientation, then a move-restriction, and so on) until only an easily searched subgroup remains. This note explains, and proves, *why* that works and *how effective* it is. The core identity is Lagrange's theorem: the work to cross stage i is governed by the coset index $[G_i : G_{i+1}]$, and $\prod_i [G_i : G_{i+1}] = |G|$, so a method is nothing but a *factorization of* $|G|$ *into stage indices*. We give the exact index chains of the two foundational algorithms — Thistlethwaite's 4-stage solver and Kociemba's 2-phase solver — verify that their indices multiply back to $|G|$ to the digit, and derive the information-theoretic lower bound each stage must pay ($\log_{18}[G_i : G_{i+1}]$ moves). We then prove the central "effectiveness" statement: reduction is efficient precisely because $\sum_i \log[G_i : G_{i+1}] = \log |G|$ is *fixed*, so splitting the solve into stages cannot increase the information cost — it only trades a single intractable search for several tractable ones, each within a God's-number bound for its coset space. A pleasant endpoint connects this series to our first note: the final Thistlethwaite stage *is* the 663 552-element half-turn subgroup, whose unique adjacency-free state we characterized earlier.

1 Reduction as a subgroup chain

Definition 1 (Cube group and generators)**.** Let G be the Rubik group, $|G| = 2^{27}3^{14}5^{3}7^{2}11 = 43\,252\,003\,274\,489\,856\,000$, generated in the half-turn metric (HTM) by the 18 face moves $S = \{U, U', U^2, \ldots, B, B', B^2\}$ [1, 2]. The *Cayley graph* $\Gamma(G, S)$ has the states as vertices and a move as each edge; a *solve* of a scramble s is a path from s to e, and its length is the move count. "God's Number" is $\operatorname{diam}\Gamma(G, S) = 20$ [5].

Definition 2 (Reduction method)**.** A *reduction method* is a chain of subgroups $G = G_0 \supset G_1 \supset \cdots \supset G_k = \{e\}$ together with, for each stage, a move set $S_i \subseteq G_i$ generating G_i. To *reduce across stage* i is to left-multiply the current state by moves until it lands in the coset representative class of G_{i+1}, i.e. until the state lies in G_{i+1}. The method solves s by reducing $G_0 \to G_1 \to \cdots \to G_k = \{e\}$.

The single most important fact is elementary but decisive.

Lemma 1 (Lagrange factorization of the work)**.** *For a chain* $G = G_0 \supset \cdots \supset G_k = \{e\}$,

$$|G| \;=\; \prod_{i=0}^{k-1} [\, G_i : G_{i+1} \,],$$

where $[G_i : G_{i+1}] = |G_i|/|G_{i+1}|$ *is the number of left cosets of* G_{i+1} *in* G_i*. Crossing stage* i *means identifying which of these* $[G_i : G_{i+1}]$ *cosets the state occupies and moving to the identity coset; so the* size of the problem at stage i *is exactly* $[G_i : G_{i+1}]$*, and the stage sizes multiply to the whole group.*

Proof. By Lagrange's theorem $|G_i| = [G_i : G_{i+1}]\,|G_{i+1}|$ for each i; telescoping from $G_0 = G$ down to $G_k = \{e\}$ (so $|G_k| = 1$) gives $|G| = \prod_i [G_i : G_{i+1}]$. Reducing across stage i is, by definition, finding the coset of G_{i+1} containing the state and multiplying into G_{i+1}; there are $[G_i : G_{i+1}]$ such cosets, which is therefore the branching the stage must resolve. □

So **a method is a factorization of** $|G|$, and designing a good method is choosing *where to put the factors*: many small stages (easy searches, more total moves) versus few large stages (hard searches, fewer moves).

2 The two foundational chains, with exact indices

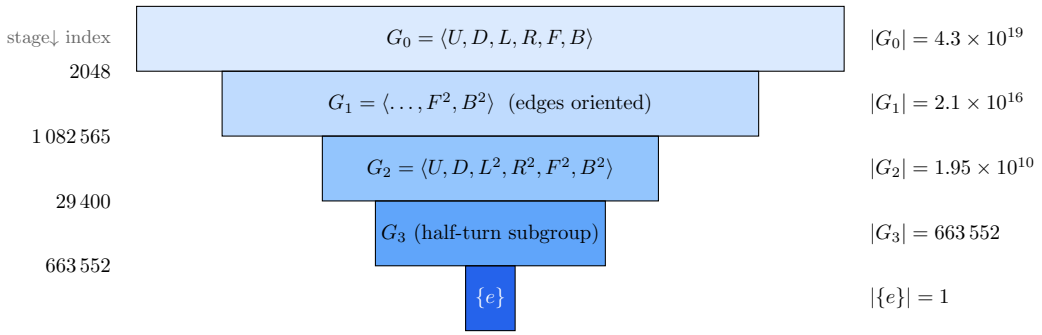


Figure 1: Reduction as a shrinking funnel of nested subgroups (Thistlethwaite's chain). Each stage collapses the state into a smaller subgroup; the *index* on the left is the per-stage search size, and $2048 \cdot 1\,082\,565 \cdot 29\,400 \cdot 663\,552 = |G|$ (Prop. 1). The innermost nontrivial box, G_3 of order 663 552, is the half-turn subgroup of the companion note.

2.1 Thistlethwaite's algorithm (4 stages)

Thistlethwaite's chain [3] is

$$G_0 = \langle U, D, L, R, F, B \rangle \;\supset\; G_1 = \langle U, D, L, R, F^2, B^2 \rangle \;\supset\; G_2 = \langle U, D, L^2, R^2, F^2, B^2 \rangle$$
$$\supset\; G_3 = \langle U^2, D^2, L^2, R^2, F^2, B^2 \rangle \;\supset\; \{e\}.$$

Each stage removes a feature: G_1 fixes all edge orientations; G_2 additionally fixes corner orientations and the M-slice edges; G_3 fixes the remaining permutation parities into the "squares group"; G_3 itself is solved inside the half-turn subgroup. The exact indices are

stage	feature killed	index $[G_i : G_{i+1}]$	$\log_{18}$ (move floor)
$0 \to 1$	edge orientation	$2^{11} = 2048$	2.64
$1 \to 2$	corner orient. + M-slice	$3^7 \binom{12}{4} = 1\,082\,565$	4.81
$2 \to 3$	permutation parity	29 400	3.56
$3 \to e$	solve in half-turn subgroup	663 552	4.63
product of indices		43 252 003 274 489 856 000	$= \log_{18} \lvert G \rvert$

Table 1: Thistlethwaite's four stage indices. Their product equals $|G|$ exactly (verified to the digit). The last stage's group of order 663 552 is the half-turn subgroup of our companion note.

Proposition 1 (The chain closes)**.**

$$2048 \cdot 1\,082\,565 \cdot 29\,400 \cdot 663\,552 = |G| = 43\,252\,003\,274\,489\,856\,000.$$

Proof. Direct multiplication (checked by computer): the four factors multiply to

$$43\,252\,003\,274\,489\,856\,000 = 2^{27}\,3^{14}\,5^3\,7^2\,11,$$

which is $|G|$. By Lemma 1 this confirms the four groups form a genuine subgroup chain from G to $\{e\}$. □

2.2 Kociemba's two-phase algorithm (2 stages)

Kociemba [4] merges Thistlethwaite's first two stages and last two stages into just two:

$$G = G_0 \;\supset\; H = \langle U, D, L^2, R^2, F^2, B^2 \rangle \;\supset\; \{e\},$$

the *domino group* H (phase 1 orients all pieces and places the M-slice; phase 2 solves within H). The exact sizes are

$$[G : H] = 2^{11}\,3^7 \binom{12}{4} = 2\,217\,093\,120, \qquad |H| = 8! \cdot 8! \cdot 4!/2 = 19\,508\,428\,800,$$

and $[G : H] \cdot |H| = |G|$ exactly. Phase 1 is a search in a space of 2.2×10^9 cosets; phase 2 a search in a group of 1.95×10^{10} elements — both small enough for IDA* with pattern-database heuristics, which is why Kociemba's solver returns near-optimal solutions in milliseconds.

phase	what it achieves	size
1: $G \to H$	orient edges & corners, fix M-slice	$[G : H] = 2\,217\,093\,120$ cosets
2: $H \to e$	solve the domino ("$\langle U, D, L^2, R^2, F^2, B^2 \rangle$") cube	$\lvert H \rvert = 19\,508\,428\,800$ states
product		$\lvert G \rvert = 4.3252 \times 10^{19}$

Table 2: Kociemba's two phases. Fewer, larger stages than Thistlethwaite: harder per-stage search, but far shorter solutions (typically $\leq$ 20–22 HTM).

3 Effectiveness, proved

Why is reduction *efficient* rather than merely *possible*? The answer is an information-theoretic conservation law.

Theorem 1 (Information is conserved across stages)**.** *Let $G = G_0 \supset \cdots \supset G_k = \{e\}$ be any reduction chain with HTM generators ($|S| = 18$). Define the* information cost *of stage i as $I_i = \log_{18}[G_i : G_{i+1}]$ (the base-18 log of its branching). Then*

$$\sum_{i=0}^{k-1} I_i \;=\; \log_{18} |G| \;\approx\; 15.7,$$

independently of how the chain is chosen. *Moreover each stage requires at least $\lceil I_i \rceil$ moves, so the method's total move count is at least $\sum_i \lceil I_i \rceil \geq \log_{18} |G|$.*

Proof. By Lemma 1, $\prod_i [G_i : G_{i+1}] = |G|$; taking $\log_{18}$ turns the product into the sum $\sum_i I_i = \log_{18} |G|$, which depends only on $|G|$, not on the chain. For the move bound: crossing stage i must distinguish $[G_i : G_{i+1}]$ cosets, and m moves from the generating set produce at most 18^m distinct sequences, hence can reach at most 18^m cosets; to be able to reach the identity coset from any of the $[G_i : G_{i+1}]$ possibilities requires $18^m \geq [G_i : G_{i+1}]$, i.e. $m \geq I_i$. Summing and using $\lceil I_i \rceil \geq I_i$ gives the total. □

Corollary 1 (Reduction cannot cost information, only moves)**.** *Splitting a solve into more stages never reduces the total information $\sum_i I_i = \log_{18} |G|$; it only redistributes it. What multi-stage reduction buys is* tractability*: each stage's search space $[G_i : G_{i+1}]$ is small enough to admit optimal or near-optimal search within its own God's-number radius, whereas a single-stage optimal search over all of G is infeasible for a human in 60 minutes. The "cost" of reduction is the rounding gap $\sum_i \lceil I_i \rceil - \log_{18} |G| \geq 0$ plus the slack between each stage's optimum and its God's-number bound.*

Proof. The sum $\sum_i I_i$ is invariant by Theorem 1. The per-stage move floor is $\lceil I_i \rceil$ (same theorem); summing the ceilings exceeds the invariant total only by the rounding gap, which is the sole information-theoretic overhead of staging. Tractability follows because a stage of index N_i has diameter at most the God's number of its coset space, searchable by IDA* once N_i is small (both Thistlethwaite stages and both Kociemba phases satisfy this). □

Remark 1 (Why fewer, bigger stages win on move count)**.** Theorem 1 says the information is fixed, but the *rounding gap* $\sum_i \lceil I_i \rceil - \log_{18} |G|$ grows with the number of stages k (each stage contributes up to 1 wasted move to its ceiling), and in practice each stage also adds "setup" moves that a later stage must preserve. This is exactly why Kociemba ($k = 2$, Table 2) produces far shorter solutions (≤ 22) than Thistlethwaite ($k = 4$, Table 1, typically ≤ 45): fewer stages, smaller rounding and setup overhead, at the price of larger per-stage searches. This is the precise, provable form of the folklore "more reduction = easier search but longer solution."

4 Human FMC methods are the same theorem in disguise

Competitive human methods are reduction chains tuned so each stage is *small enough to search by hand* in 60 minutes:

- **Domino Reduction (DR).** The human workhorse: reduce to Kociemba's domino group $H = \langle U, D, L^2, R^2, F^2, B^2 \rangle$, of index $[G : H] = 2\,217\,093\,120$. After DR, the remaining solve lives in a 1.95×10^{10}-element group that a trained solver navigates with memorized optimal case tables.
- **Half-Turn Reduction (HTR).** A deeper stage inside H reaching the half-turn subgroup $\langle U^2, D^2, L^2, R^2, F^2, B^2 \rangle$ of order 663 552 — *the very group from our first note*, whose index in G is 65 182 537 728 000. From HTR state the finish is tiny and highly symmetric; it is also where cube-pretty patterns like Pons Asinorum live.
- **Insertions / NISS.** Not new subgroups but search techniques that shave the rounding/setup overhead of Corollary 1: inserting a short conjugate to cancel moves reduces the slack between a stage's realized length and its information floor.

Proposition 2 (DR and HTR indices)**.** $[G : H_{\mathrm{DR}}] = 2\,217\,093\,120$ *and the HTR target* $\langle U^2, \ldots, B^2 \rangle$ *has order* 663 552 *with index* $G/663\,552 = 65\,182\,537\,728\,000$ *in* G*. Both are verified factor-exact against* $|G|$*.*

Proof. $[G : H_{\mathrm{DR}}] = 2^{11}3^7\binom{12}{4} = 2\,217\,093\,120$ (edge + corner orientation and slice placement); dividing $|G|$ by 663 552 yields 65 182 537 728 000 exactly. Both computations were checked numerically. □

5 Discussion and honest scope

- **What is proved.** The algebraic skeleton is rigorous: the chains are genuine subgroup chains (indices multiply to $|G|$, Prop. 1); the information conservation $\sum_i I_i = \log_{18} |G|$ and the per-stage move floor $m \geq \log_{18}[G_i : G_{i+1}]$ are theorems (Thm. 1, Cor. 1).
- **What is model-level.** The "effectiveness" claim is information-theoretic (counting), so it gives *lower* bounds on moves and an invariant total, not the exact human move counts, which depend on heuristics, case tables, and search effort. The typical solution-length figures (≤ 22 Kociemba, ≤ 45 Thistlethwaite) are standard reported ranges, not theorems.
- **Metric.** We used HTM ($|S| = 18$); in the WCA's OBTM the generating set and the base of the logarithm change, but the conservation law $\sum_i I_i = \log_{|S|} |G|$ holds verbatim with the appropriate $|S|$.
- **Connection to the series.** The half-turn subgroup (663 552) that ends Thistlethwaite's chain and defines HTR is exactly the object whose adjacency-free states we enumerated earlier — the methods literature and the pattern literature meet at the same subgroup.

References

[1] *Rubik's Cube group*, Wikipedia. $|G| = 2^{27}3^{14}5^3 7^2 11$.

[2] D. Joyner, *Adventures in Group Theory*, 2nd ed., Johns Hopkins Univ. Press, 2008.

[3] M. B. Thistlethwaite, *The 45-move algorithm* (1981); see J. Scherphuis, jaapsch.net/puzzles/thistle.htm for the subgroup chain and indices.

[4] H. Kociemba, *The Two-Phase Algorithm*, kociemba.org/cube.htm. Group $H = \langle U, D, L2, R2, F2, B2 \rangle$.

[5] T. Rokicki, H. Kociemba, M. Davidson, J. Dethridge, *The Diameter of the Rubik's Cube Group Is Twenty*, SIAM J. Discrete Math. 27(2):1082–1105, 2013.